TWO-HOUR TUTORIAL

Recurring decimals and exact fractions

MYP Year 5 eAssessment preparation | Standard mathematics, required for Extended

The central idea

A recurring decimal is a rational number in disguise. The dots tell you which digits repeat. Your task is to read that notation exactly, convert to a fraction in simplest form, and explain why the method works.

Name: _____ Date: _____

Your 120-minute route

Minutes	Focus and activity	Page
0–8	Diagnose starting points: D1–D6	2
8–20	Tutorial: notation, terminating and pure recurring	3
20–34	Apply the shift method: A1–A6	4
34–48	Mixed recurring decimals: tutorial + M1–M4	5–6
48–60	Both directions, and when a decimal terminates: P1–P4	7
60–82	Investigate the nines family: B1–B5	8
82–98	Use an exact value in context: T1–T3	9
98–113	Independent checkpoint: C1–C4	10
113–120	Exit questions and feedback: X1–X3	11

What you will be able to do

- Read IB recurring-decimal notation, including a block and a non-repeating start.
- Convert terminating, pure recurring and mixed recurring decimals to simplified fractions.
- Decide from a fraction in lowest terms whether its decimal terminates or recurs.
- Describe, test, verify and justify the rule “the repeating block over a string of nines”.
- Keep an exact value when a rounded decimal would damage a later calculation.

CORE = this whole 120-minute route. It is in the standard skills framework and is also required for extended mathematics. **ENRICHMENT** = the general proof and a three-digit mixed decimal on pages 12–13. Page 14 is a reference sheet. These are outside the 120 minutes.

This is an intensive revision lesson: you should already know equivalent fractions, simplifying, and multiplication by powers of 10. All questions are original. This is an independent Palmer Grove resource, not an official IB paper.

0–8 minutes | Core

Start with what you know

6 diagnostic marks | First 5 minutes: independent. Then compare and explain.

D1 Write 0.375 as a fraction in simplest form. [1]

D2 In $0.\dot{2}\dot{7}$, which digits repeat, and in what order? [1]

D3 State whether $0.\dot{4}$ is rational or irrational. Give one reason. [1]

D4 Write the terminating decimal 0.16 as a fraction in simplest form. [1]

D5 In $0.1\dot{6}$, which digits repeat? [1]

D6 A calculator shows 0.181818 . . . How should this be written with recurring dots? [1]

Prepare to communicate on screen

On screen, place a dot over the **first and the last** digit of the repeating block. One repeating digit needs one dot: $0.\dot{3}$. A block of two digits needs two dots: $0.\dot{2}\dot{7}$ means 0.272727 . . ., not 0.2777 . . .. Check the equation editor before you leave the item. Final answers should be simplified fractions, not calculator strings.

Quick self-check: I am most secure with _____
I need to revisit _____

8–20 minutes | Core

Read the dots, then shift

Teacher models; you explain each change of form.

Three forms, one idea

A **terminating** decimal ends. A **recurring** decimal repeats a block forever. Both are rational: they can be written as p/q with integers p and q, and $q \neq 0$. A decimal that neither ends nor repeats is irrational.

eAssessment notation. Dots sit over the first and last digits of the repeating block.

$$0.\dot{3} = 0.3333\ldots, \qquad 0.\dot{1}\dot{8} = 0.181818\ldots, \qquad 0.1\dot{6} = 0.16666\ldots$$

The last of these is *mixed*: 1 does not repeat, and 6 does.

Worked example 1 | A terminating decimal uses place value

$$0.375 = \frac{375}{1000} = \frac{3}{8}.$$

Divide numerator and denominator by their highest common factor, 125.

Worked example 2 | One repeating digit

Let $x = 0.\dot{7} = 0.7777\ldots$

Multiply by 10, because one digit repeats:

$$10x = 7.7777\ldots$$

Subtract the original. The repeating tails cancel.

$$10x - x = 7, \qquad 9x = 7, \qquad x = \frac{7}{9}.$$

The denominator is a single 9, one nine for one repeating digit.

Worked example 3 | A two-digit block

Let $x = 0.\dot{2}\dot{7} = 0.272727\ldots$

Multiply by 100, because the block has two digits:

$$100x = 27.272727\ldots$$

$$100x - x = 27, \qquad 99x = 27, \qquad x = \frac{27}{99} = \frac{3}{11}.$$

Two repeating digits give two nines. Then simplify.

Explain aloud: why is the multiplier 100 here, but 10 in example 2?

- $0.\dot{3}$ is not 3/10. The digit 3 never stops, so the denominator is 9, not 10.
- $0.\dot{1}\dot{6}$ and $0.1\dot{6}$ are different numbers. Read the dots before you multiply.
- An unsimplified fraction such as 27/99 is not the finished answer unless the question says otherwise.

Teacher prompts: "How many digits are in the block?" "What power of 10 shifts exactly one block?" "What cancels when you subtract?"

20–34 minutes | Core

Apply the shift and challenge an error

11 practice marks | The repeating part starts immediately after the decimal point.

A1 Write $0.\dot{4}$ as a fraction in simplest form. [1]

A2 Write $0.\dot{1}\dot{8}$ as a fraction in simplest form. Show the subtraction. [2]

A3 Write $0.\dot{0}6$ as a fraction in simplest form. The first repeating digit is 0. [2]

A4 Write $2.\dot{4}$ as an improper fraction in simplest form. [2]

A5 Show that $0.\dot{2}\dot{7} = \dfrac{3}{11}$. [2]

A6 A pupil claims that $0.\dot{3} = \dfrac{3}{10}$ because “there is one decimal place”. Explain the mistake and give the correct fraction. [2]

34–48 minutes | Core

Mixed recurring decimals

Some digits happen once. Only the dotted block repeats.

Shift twice, then subtract

If digits occur *before* the repeating block, one multiplication is not enough.

Let $x = 0.2\dot{5} = 0.25555\ldots$

$$\begin{aligned} 10x &= 2.5555\ldots \\ 100x &= 25.5555\ldots \\ 100x - 10x &= 23 \\ 90x &= 23 \\ x &= \frac{23}{90}. \end{aligned}$$

The 10 moves the non-repeating digit to the left of the decimal point. The 100 moves one further place, through the first repeating digit. Their difference, 90, clears the tail.

Worked example 4 | A familiar fraction, read carefully

$0.\dot{1}6 = 0.16666\ldots$, not $0.161616\ldots$

$$\begin{aligned} 10x &= 1.6666\ldots \\ 100x &= 16.6666\ldots \\ 90x &= 15 \\ x &= \frac{15}{90} = \frac{1}{6}. \end{aligned}$$

Compare: $0.\dot{1}\dot{6} = 0.161616\ldots = \frac{16}{99}$. Same digits, different dots, different value.

A check that does not need a calculator. Convert the fraction back by division for one or two places, and confirm that the repeating digit matches.

11 practice marks

M1 Write $0.2\dot{7}$ as a fraction in simplest form. Show both multiplications. [3]

M2 Show that $0.\dot{1}6 = \frac{1}{6}$. [2]

M3 Write $0.12\dot{3}$ as a fraction in simplest form. [3]

M4 Write both numbers as simplified fractions, and state which is larger.

$$0.\dot{1}\dot{6} \qquad \text{and} \qquad 0.1\dot{6}$$

[3]

If you misread $0.1\dot{6}$ as $0.\dot{1}\dot{6}$, every later line can be correct algebra of the wrong number. Say the decimal aloud before you choose the multipliers: “zero point one, six recurring” or “zero point one-six recurring”.

48–60 minutes | Core

Both directions

The eAssessment may start from either form.

Fraction to decimal. Divide the numerator by the denominator. If a remainder repeats, the digits repeat.

$$\frac{5}{6} = 0.8333\ldots = 0.8\dot{3}, \qquad \frac{2}{11} = 0.181818\ldots = 0.\dot{1}\dot{8}.$$

Terminating or recurring?

Write the fraction in lowest terms. Look only at the denominator.

- If its prime factors are **only** 2s and/or 5s, the decimal terminates.
- If any other prime appears, the decimal recurs.

$\frac{3}{8}$ terminates because $8 = 2^3$. $\frac{5}{12}$ recurs because $12 = 2^2 \times 3$ and the factor 3 remains after simplifying.

P1 Write $\frac{5}{6}$ using recurring-dot notation. [1]

P2 Write $\frac{2}{11}$ using recurring-dot notation. [1]

P3 For each fraction, state **terminating** or **recurring**. Use the prime factors of the denominator in lowest terms. You do not need the decimal itself.

a) $\frac{9}{40}$

b) $\frac{7}{15}$

c) $\frac{5}{12}$

d) $\frac{3}{8}$

[4]

P4 Write $\frac{9}{40}$ as a decimal, and state why your answer agrees with P3a. [2]

60–82 minutes | Criterion B

Investigate the nines family

Cases, pattern, rule, verify, justify. | 14 marks. | A correct table is not yet a rule.

A purely recurring decimal has no non-repeating digits after the decimal point. This investigation is about that family only.

B1 Write each fraction with recurring dots. [2]

$$\frac{1}{9}, \quad \frac{2}{9}, \quad \frac{4}{9}, \quad \frac{8}{9}$$

B2 Write $\frac{12}{99}$ and $\frac{7}{99}$ with recurring dots. [2]

B3 Describe the pattern. For a purely recurring decimal, what stays the same about the denominator, and what changes with the number of digits in the block? [2]

B4 State a general rule. If the repeating block is a k-digit whole number N (a leading zero in the block still counts as a digit), then

$$0.\overline{N} = \frac{N}{\square}.$$

Complete the denominator in terms of k, and give the rule in a sentence. [3]

B5 Verify and justify.

a) Use your rule to write $0.\dot{1}4\dot{2}$ as a fraction. Check whether it simplifies. [2]

b) Let $x = 0.\dot{1}4\dot{2}$. Show algebraically why multiplying by 10^3 and subtracting x produces that same fraction. [3]

Communication: define x, show the subtraction, and finish with a sentence that links the algebra to the rule.

82–98 minutes | Criterion D

Keep the exact reading

A sensor display has already rounded. The exact value may still be available from the repeating form.

A water-quality probe reports a concentration as a recurring decimal, in milligrams per litre.

T1 The display shows $0.1\dot{6}$ mg/L. Write this concentration as a fraction in simplest form. [2]

T2 An operator records 0.17 mg/L instead. Find the difference

$$0.17 - 0.1\dot{6}$$

as a fraction. State whether the recorded value is too high or too low. [3]

T3 The sample volume is 90 litres, and the mass of the substance, in milligrams, is concentration times volume. Using the exact concentration from T1, find the mass. Then explain why replacing $0.1\dot{6}$ by 0.17 before this multiplication changes the mass. [3]

Prepare to communicate on screen

In an on-screen context item, the final line should answer the situation, not only the algebra: too high or too low, and by what exact amount. Keep the fraction until the last step.

98–113 minutes

Independent checkpoint

10 marks | 15 minutes | Close the reference sheet. Show enough working to recover a mark after a slip.

C1 Write $0.\dot{4}\dot{5}$ as a fraction in simplest form. [2]

C2 Write $0.4\dot{5}$ as a fraction in simplest form. [3]

C3 Explain why $0.\dot{2}\dot{7}$ is rational, while a decimal that does not terminate and does not recur is not rational. [2]

C4 Use the shift method to show that $0.\dot{9} = 1$. [3]

One thing I must not confuse in the next paper: _____

113–120 minutes

Exit check and next step

5 marks | 3 minutes to answer; 4 minutes to review and act on feedback.

X1 Write $0.\dot{6}$ as a fraction in simplest form. [1]

X2 Write $3.1\dot{2}\dot{3}$ as an improper fraction in simplest form. [2]

X3 A fraction has been simplified from $90x = 15$ to $\frac{1}{6}$. Why is the second form the better final answer? [2]

Record evidence, not just a score

Skill	Evidence	Next action
Read the dots correctly	D2, D5, M4	
Pure recurring, then simplify	A2, A5, C1	
Mixed recurring	M1, C2	
A rule and its justification	B4, B5	
Exact value in context	T2, T3	

My most useful correction today: _____

One question I will redo from a blank page tomorrow: _____

After the lesson

Redo one missed core question without looking at the solution. If that is secure, complete E1 or E2 on the next two pages and write the justification in full sentences. The mark scheme includes every question.

Optional | Outside the 120 minutes

Challenge: why the nines appear

Enrichment | Extended reasoning. The algebra is the same method, used generally.

Let the repeating block have k digits, and let N be the integer formed by that block. For a purely recurring decimal,

$$x = \frac{N}{10^k} + \frac{N}{10^{2k}} + \frac{N}{10^{3k}} + \cdots$$

This is an infinite geometric series with first term $N/10^k$ and common ratio $1/10^k$. Its sum is

$$x = \frac{N/10^k}{1 - 1/10^k} = \frac{N}{10^k - 1}.$$

$10^k - 1$ is the number made of k nines. That is why the shortcut and the shift method agree.

E1 Use the shift method, not the series, to prove that a purely recurring decimal with a k-digit block N equals $N/(10^k - 1)$. State the multiplier you use. [4]

E2 Write $0.1\dot{4}\dot{2}$ as a fraction in simplest form. This is mixed: one non-repeating digit, then a two-digit block. [4]

Optional | Outside the 120 minutes

Challenge: order different forms

Do not convert everything to a rounded decimal and then guess.

E3 Arrange in ascending order. Show the exact form you used for each number.

$$\frac{5}{8}, \qquad 0.6\dot{7}, \qquad \frac{2}{3}, \qquad \frac{7}{11}$$

[5]

E4 $0.\dot{9}$ and 1 are equal, not "infinitely close". Using your result from C4, explain what is wrong with the sentence "$0.\dot{9}$ is the number just before 1". [3]

Reference | Keep for retrieval

One page of useful connections

Use during the tutorial. Close it during the independent checkpoint.

What you see	What to do
0.375	Place value: 375/1000, then simplify.
$0.\dot{a}$	One dot: multiply by 10, subtract, denominator 9.
$0.\dot{a}\dot{b}$	Dots on first and last: the block is ab. Multiply by 100.
$0.a\dot{b}$	Only b repeats. Shift by 10 and by 100, then subtract.
Block of k digits, nothing before it	Denominator $10^k - 1$, that is k nines.
Lowest-terms denominator	Only primes 2 and/or 5: terminating. Any other prime: recurring.
$0.\dot{9}$	Equals 1. The shift gives $9x = 9$.

Related ideas already in the number framework

- Rational numbers are exactly the terminating and recurring decimals.
- $\sqrt{49} = 7$ is rational. $\sqrt{12}$ is irrational: its decimal neither terminates nor recurs.
- Standard form does not replace an exact fraction. $1/6 = 1.\dot{6} \times 10^{-1}$ still hides a repeating digit; $\frac{1}{6}$ does not.
- Simplify by the highest common factor at the end, or earlier if the numbers are awkward.

Make an explanation assessable

Write: give the simplest fraction. **Show that:** reach the given result by a valid method. **Verify:** check a new case independently. **Justify:** explain why the multiplier clears the repeating tail. **In context:** say what the exact difference means for the measurement.